

Why Non-Fire Surfaces Are Detected as Fire in Airborne LWIR Imagery

Physical origins of false-positive fire detections in single-band LWIR airborne imagery

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Abstract

Long-wave infrared (LWIR, 8–14 μm) cameras carried on firefighting aircraft measure the radiance emitted by the ground surface, not its temperature. Using the radiative transfer equation, the surface energy balance, the Dozier sub-pixel mixing model and laboratory spectral-emissivity data, this note explains why solar-heated natural surfaces — especially bare rock on sun-facing slopes — produce LWIR radiance that cannot be separated from that of a small active fire. Five independent factors are identified and quantified: the surface energy balance, solar illumination geometry, the weak temperature response of the Planck function in the LWIR, the spectral emissivity of silicate minerals, and sensor and processing limitations. The implications for designing a fire-detection algorithm with a single-band LWIR sensor are then set out.

1. Introduction

Automated fire detection from airborne thermal imagery underpins products such as fire-perimeter maps and flight-crew situational awareness. The review by Allison et al. [8] notes the principal advantage of thermal imaging: smoke is nearly transparent at LWIR wavelengths, so the active fire edge is visible through it. The same review, and the operational experience of the FireMapper programme [9], also stress that thermal sensors, particularly in daytime flights, are prone to false-positive detections on warm non-fire surfaces.

Mature satellite algorithms such as MODIS [4] and VIIRS [5] solve this by combining a mid-wave infrared band (MWIR, $\sim 4 \mu\text{m}$) with an LWIR band ($\sim 11 \mu\text{m}$). When a sensor has only a single LWIR band, that discriminator is unavailable, and a precise understanding of the physics is required to design substitutes. This note presents that physics step by step.

2. Physical basis: the sensor sees radiance, not temperature

Neglecting the atmosphere, the spectral radiance reaching an LWIR pixel from a surface of emissivity ϵ and temperature T is [1, 7]:

$$L(\lambda) = \epsilon(\lambda) \cdot B(\lambda, T) + [1 - \epsilon(\lambda)] \cdot L_{\downarrow}(\lambda)$$

where B is the Planck function and $L_{\downarrow}$ is the downwelling sky radiance. The Planck function itself is:

$$B(\lambda, T) = (2hc^2 / \lambda^5) \cdot 1 / [\exp(hc / \lambda kT) - 1]$$

A detection algorithm converts L to a brightness temperature assuming a fixed ϵ , then compares it either with an absolute threshold or with the statistics of neighbouring pixels (the contextual approach [4]). Anything that raises L without a fire present produces a false detection. The five factors below do exactly that.

3. Factors producing false detections

3.1 Surface energy balance and solar heating of rock

Surface temperature is set by the energy balance [2]:

$$(1 - \alpha) \cdot S + L\downarrow - \varepsilon \sigma T^4 - H - LE - G = 0$$

where α is albedo, S the incoming solar irradiance, H the sensible heat flux, LE the latent heat flux and G the ground heat flux. Bare rock differs from vegetation in three terms. Its albedo is low, so it absorbs a larger share of S . Its latent heat flux is essentially zero because there is no water to evaporate, whereas vegetation cools itself by transpiration. And the thermal inertia of its surface layer is lower than that of moist soil or a plant canopy, so it warms faster.

The practical result: on a summer afternoon, rock surfaces reach 330–345 K while surrounding vegetation stays near 300–305 K. A 30–40 K contrast with the background is precisely the signal that contextual tests (comparison of a pixel with the mean and standard deviation of its neighbours) look for when identifying fire [4].

3.2 Solar illumination geometry and slope

The direct solar irradiance received by a surface scales with the cosine of the incidence angle [11]:

$$\cos \theta_i = \cos \theta_s \cdot \cos \beta + \sin \theta_s \cdot \sin \beta \cdot \cos(\varphi_s - \varphi)$$

where θ_s is the solar zenith angle, φ_s the solar azimuth, β the slope and φ the aspect. Figure 1 evaluates this for a 30° slope at 15:00 (sun elevation 50°, azimuth 225°): a south-west-facing slope receives about 29% more energy than flat ground and roughly three times as much as a north-east-facing slope.

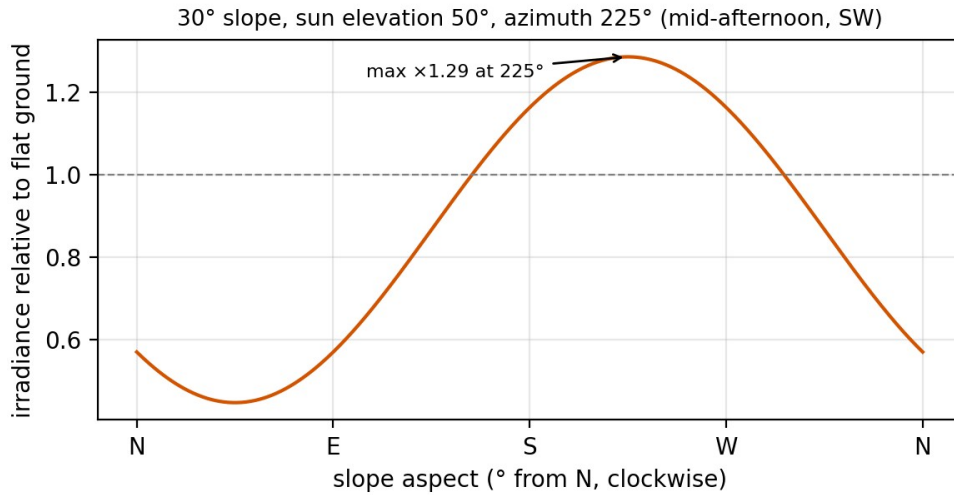


Figure 1. Relative solar irradiance on a 30° slope as a function of aspect, for an afternoon sun. Computed from the incidence-angle relation [11].

Not all rock therefore becomes hot; rock with a particular geometry does. This explains why false detections appear as isolated patches on specific slopes, and why a solar-radiation model run over a DEM can predict where those patches will be.

3.3 Weak Planck response in the LWIR and the sub-pixel problem

This is the most fundamental limitation. Dozier [3] showed that the radiance of a pixel in which a fraction p burns at temperature T_f while the remainder sits at background temperature T_b is a linear mixture:

$$L_{\text{pixel}}(\lambda) = p \cdot B(\lambda, T_f) + (1 - p) \cdot B(\lambda, T_b)$$

The problem lies in how B depends on temperature at different wavelengths. Figure 2 shows that at $4 \mu\text{m}$ (the Wien regime) radiance grows exponentially with temperature, whereas at $10 \mu\text{m}$, for terrestrial temperatures, the function approaches the Rayleigh–Jeans regime and its dependence on T is nearly linear. In numbers: raising the temperature from 300 K to 340 K multiplies radiance by about 4 at $4 \mu\text{m}$ but only by 1.8 at $10 \mu\text{m}$.

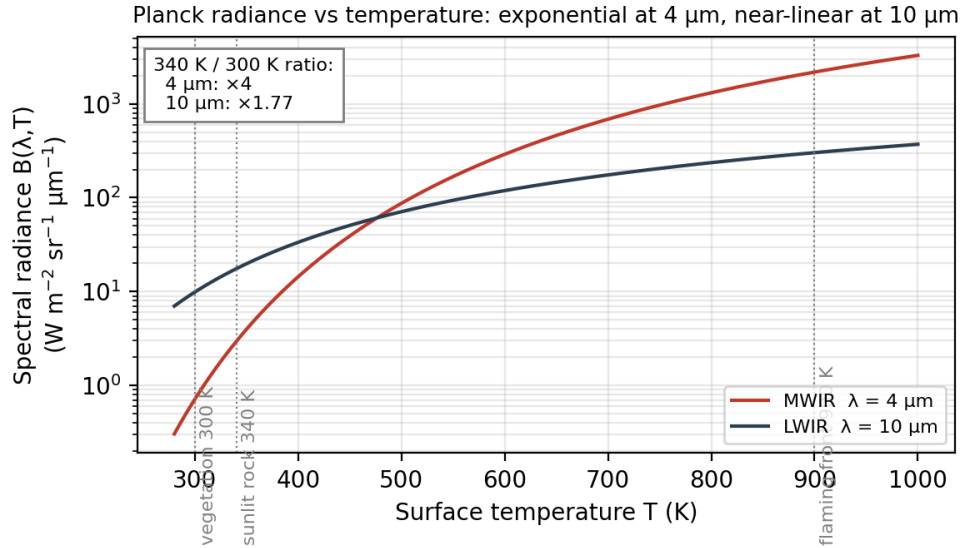


Figure 2. Planck spectral radiance versus temperature at $4 \mu\text{m}$ and $10 \mu\text{m}$ (logarithmic vertical axis).

The consequence is shown in Figure 3. At $10 \mu\text{m}$, a pixel with 2% of its area burning at 900 K has a radiance of about $15.8 \text{ W} \cdot \text{m}^{-2} \cdot \text{sr}^{-1} \cdot \mu\text{m}^{-1}$, whereas a 340 K rock filling the whole pixel gives about 17.6 — the rock is brighter than the small fire. The crossover occurs at roughly 2.6% of the pixel. At $4 \mu\text{m}$ the same 2% fire gives 44 against 2.9 for the rock, fifteen times brighter, and the crossover falls to 0.1% of the pixel.

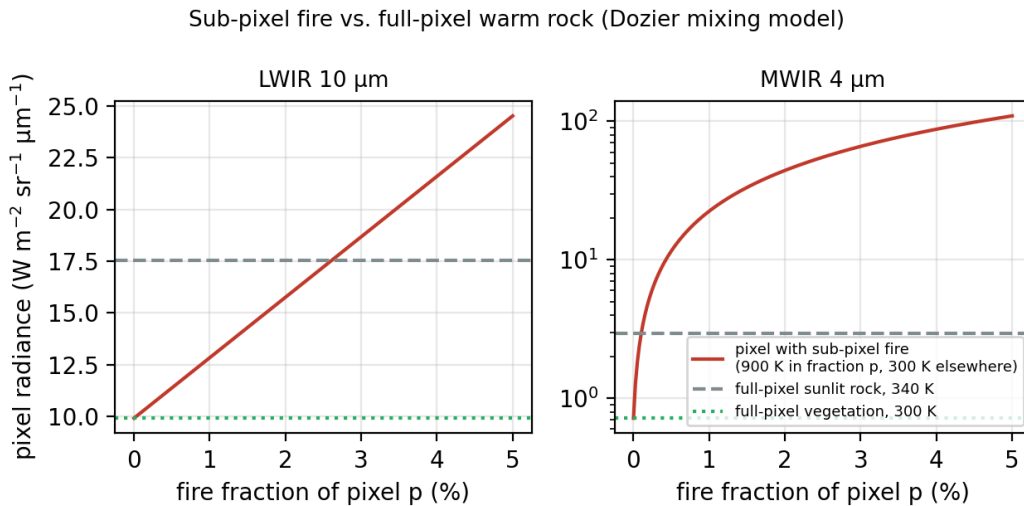


Figure 3. Dozier sub-pixel mixing model [3]: pixel radiance versus fire fraction compared with a full-pixel warm rock. Left: LWIR, linear axis; right: MWIR, logarithmic axis.

This is exactly why the MODIS [4] and VIIRS [5] algorithms perform primary detection in the $4 \mu\text{m}$ band and use the $11 \mu\text{m}$ band only to reject false alarms (the $T_4 - T_{11}$ difference test). Wooster et al. [10] built the retrieval of fire radiative power (FRP) on the same MWIR property. In the LWIR the contrast is so weak

that the band serves only for rejection, never for primary detection. With a single-band LWIR camera the discriminator is simply absent: a warm full-pixel surface and a small sub-pixel fire can return exactly the same number.

3.4 Spectral emissivity of silicate rocks

Laboratory measurements by Salisbury and D'Aria [6] show that silicate minerals (quartz, feldspar, and hence granite and sandstone) exhibit reststrahlen bands in the 8–9.5 μm region, caused by the Si–O stretching vibration, in which emissivity drops markedly (to about 0.7 or lower for pure quartz). Green vegetation, by contrast, has an emissivity near 0.98 across the entire 8–14 μm window (Figure 4).

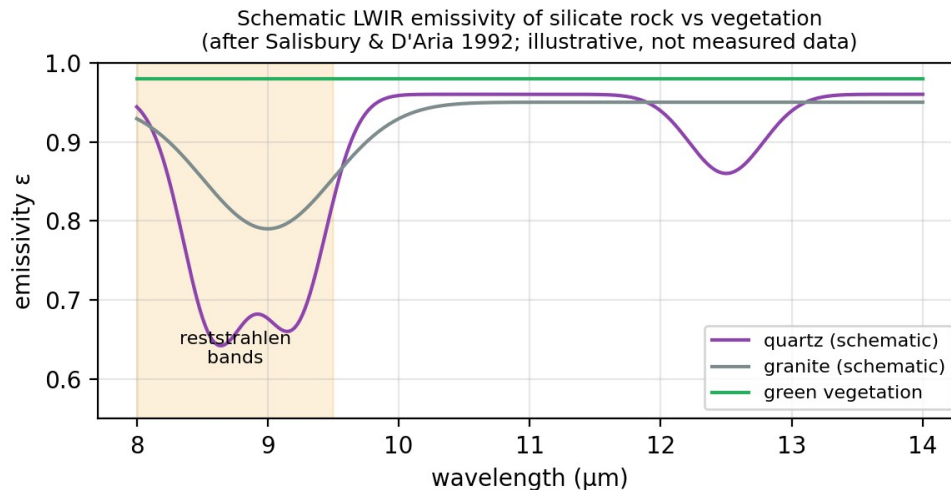


Figure 4. Schematic LWIR spectral emissivity of silicate rock versus vegetation, based on the behaviour reported in [6]. The curves are illustrative, not measured data.

If the algorithm assumes a constant emissivity (say 0.95), the temperature computed for rock is in error by several kelvin. The sign and size of the error depend on the camera's bandpass and on sky radiance, but in any case no consistent temperature map results, and a threshold that is correct for vegetation is wrong for rock. Multi-band sensors resolve this with temperature–emissivity separation (TES) [7]; with a single band the separation is mathematically impossible (one equation, two unknowns). Emissivity also decreases with increasing view angle [12], an effect that is more pronounced for smooth rock surfaces viewed obliquely.

3.5 Sensor and processing limitations

Three hardware and software issues amplify the factors above [1, 8, 9].

Saturation. The uncooled microbolometer detectors used in most airborne LWIR cameras have a limited dynamic range (typically up to a few hundred $^{\circ}\text{C}$). Over flaming combustion the pixel pins at the ceiling and intensity information is lost. Intensity therefore no longer separates fire from hot rock; all that remains is whether a threshold was crossed or not. Riggan et al. [9] designed FireMapper with an extended dynamic range for precisely this reason.

Automatic gain control (AGC). If the camera output is not radiometric and each frame is normalised to its own minimum and maximum, then in a frame with little active fire the warmest natural surface is pushed to the top of the scale and rendered red in the colour map. Red then carries no absolute temperature meaning, and comparisons between frames or passes are invalid.

View angle and atmosphere. In an oblique view the atmospheric path is longer and water-vapour absorption in the LWIR reduces the radiance received. If a uniform atmospheric correction is applied, near pixels appear warmer than far ones. In addition, any georeferencing error maps a genuine ground detection to the wrong location, producing what looks like a detection far from the fire when it is in fact a spot fire or residual heat.

4. Implications for algorithm design with single-band LWIR

Summarising Section 3: bare rock is detected as fire because (1) the energy balance makes it 30–40 K warmer than its background, (2) slope and sun geometry intensify that heating at specific locations, (3) in the LWIR the Planck response is so weak that a warm full-pixel surface cannot be separated from a sub-pixel fire, (4) silicate emissivity corrupts the radiance-to-temperature conversion, and (5) saturation and AGC destroy the intensity information that might otherwise have helped.

Since spectral discrimination is unavailable, discrimination must come from three other sources:

- Location: fuel and land-cover masks (e.g. the non-burnable LANDFIRE classes) together with a solar-radiation model over the DEM (Section 3.2) that predicts natural hot spots before the flight.
- Time: persistence across successive passes. Fire spreads, changes shape and remains hot after sunset; rock is static and cools at a rate that can be modelled (Section 3.1). Sensor artefacts are fixed in frame coordinates, not ground coordinates.
- Shape and texture: an active fire edge is linear and continuous and consistent with wind and slope; hot rock is patchy and follows geology. These features are learnable by a computer-vision model trained with real negative examples.

Night flights remove factors 3.1 and 3.2 almost entirely and provide the best validation data [8]. Contextual tests with a dynamic threshold (dependent on time of day, aspect and local background temperature) rather than a fixed one reduce sensitivity to factor 3.1. And if a hardware upgrade is possible, adding an MWIR channel removes factor 3.3, the most fundamental limitation, enabling MODIS/VIIRS-style dual-band tests [4, 5] and FRP retrieval [10].

5. Conclusion

False detection of warm surfaces in the LWIR is not a simple software bug but the joint result of several physical facts, the most important being the shape of the Planck function between 8 and 14 μm . Any single-band LWIR algorithm must accept this limitation, obtain its discrimination from location, time and shape, and report a confidence level to the operational user that reflects the conditions (day or night, sun-facing slope or not, first pass or repeat).

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Note: Figures 1–3 were computed directly from the relations cited; Figure 4 is an explanatory schematic. None reproduces a figure from the referenced works.